

The Archetype "Growth and Underinvestment" in the Details Manufacturing Market

Dreamland Incorporation

Independent Research Division

“Understanding systems before we change them”

Abstract

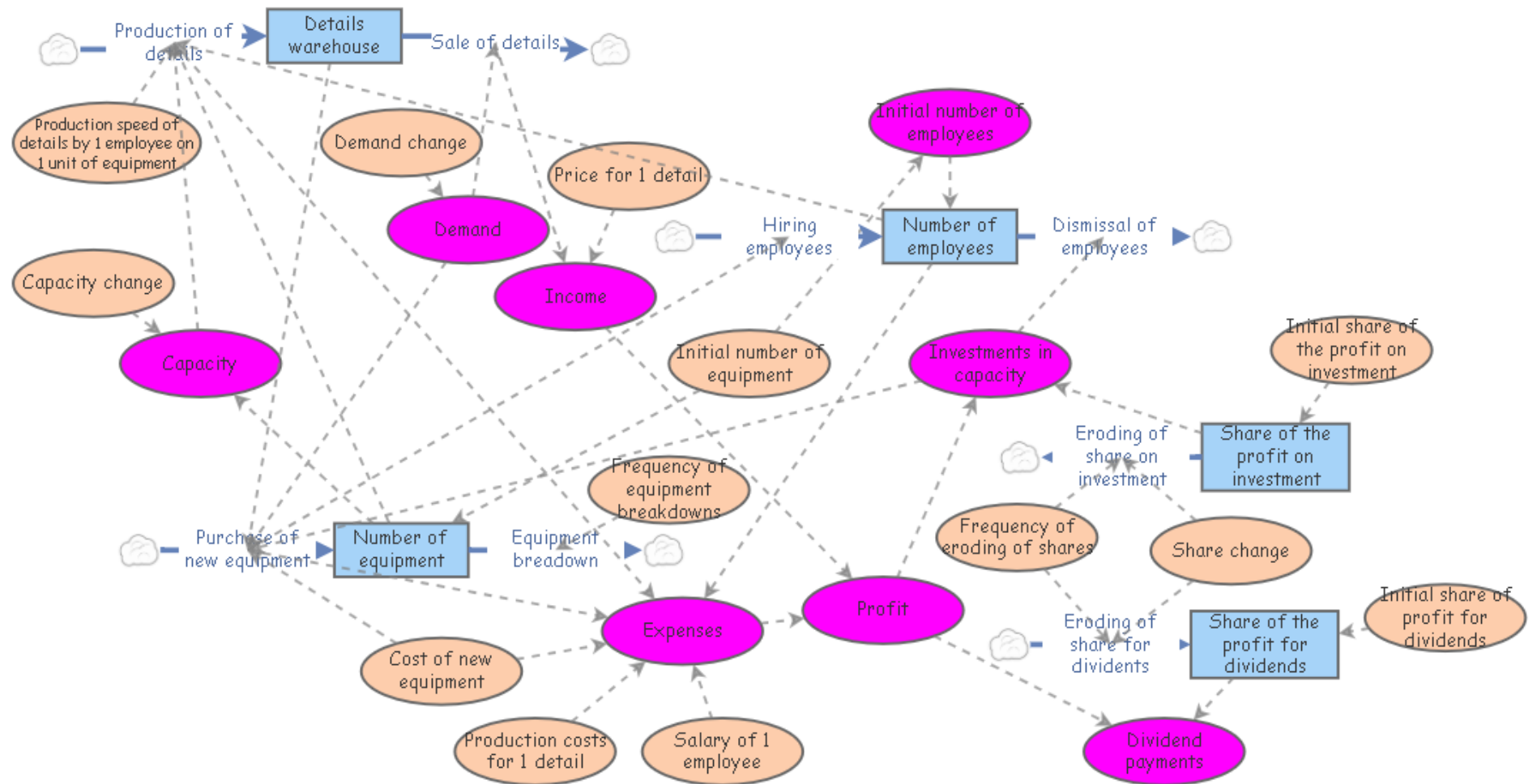
The paper explores the archetype of "Growth and Underinvestment" using the example of a manufacturing company. Three scenarios are considered, with different share of profit erosion in favor of dividends (14%, 7%, and 0%). It is shown that any positive erosion eventually leads to collapse: investments decrease, production capacity shrinks, and sales fall to zero. The only sustainable scenario is zero erosion, where the company maintains a balance between investments and dividends, ensuring long-term growth.

1. Introduction

The "Growth and Underinvestment" archetype describes a system where the current production capacity limits growth. Growth approaches a limit that can be pushed outward if the company makes sufficient and timely investments. However, if investments are insufficient or delayed, the efforts are wasted.

Often, to justify underinvestment, key goals or performance standards are lowered. This triggers a self-fulfilling prophecy: lower goals lead to lower expectations, which are then reinforced by the poor performance caused by underinvestment. A diagram of the archetype is shown in Figure 1.

Figure 2. Model diagram in Insight Maker



Model details:

Initially, the company's capacity is 30% (6 units of equipment). Purchasing new equipment increases capacity by 5%. The capacity percentage directly affects product quality and quantity. The more defective details, the lower sales, and consequently the lower profit.

Initial demand is zero but grows over time, requiring the company to increase capacity. The decision to purchase new equipment is based on the share of profit on investment. If that share exceeds the cost of new equipment, it is bought. If all profit goes to dividends, the company remains without new equipment.

Profit is calculated as revenue from details sales minus expenses (equipment purchase, production costs, employee salaries). The initial profit shares are 70% for investment and 30% for dividends. Every 12 months, these shares are eroded by 14%, 7%, or 0% (depending on the scenario).

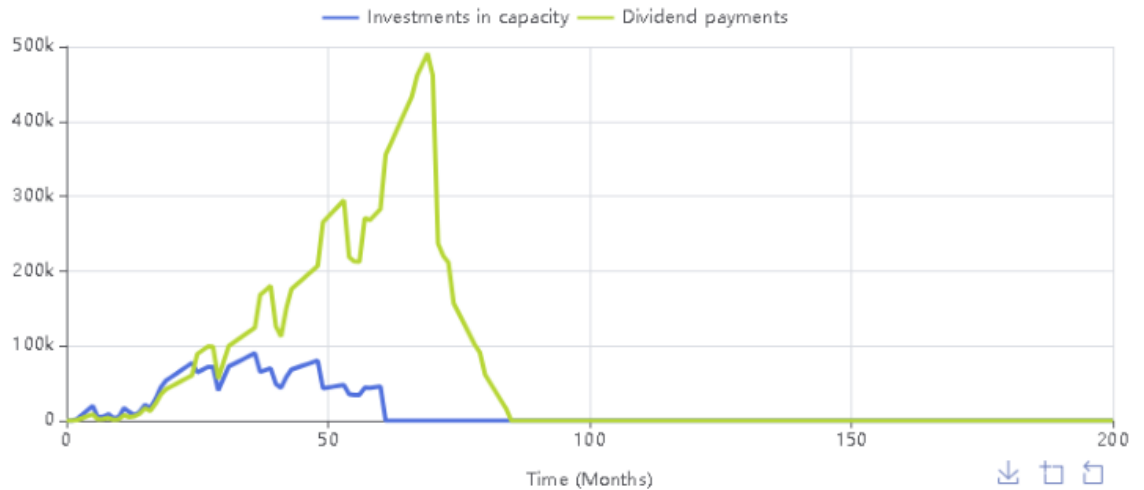
2.1. Experiment with erosion rate = 14%

The results of calculations of the model in Insight Maker with an erosion of shares equal to 14% for the first 15 steps are shown in Figure 3. Graphs of the dynamics of investments in capacity and dividend payments (Figure 4), dynamics of capacity (Figure 5), dynamics of warehouse, sales and demand (Figure 6) for the entire time are also presented (200 steps).

Figure 3. Model results with erosion rate = 14%

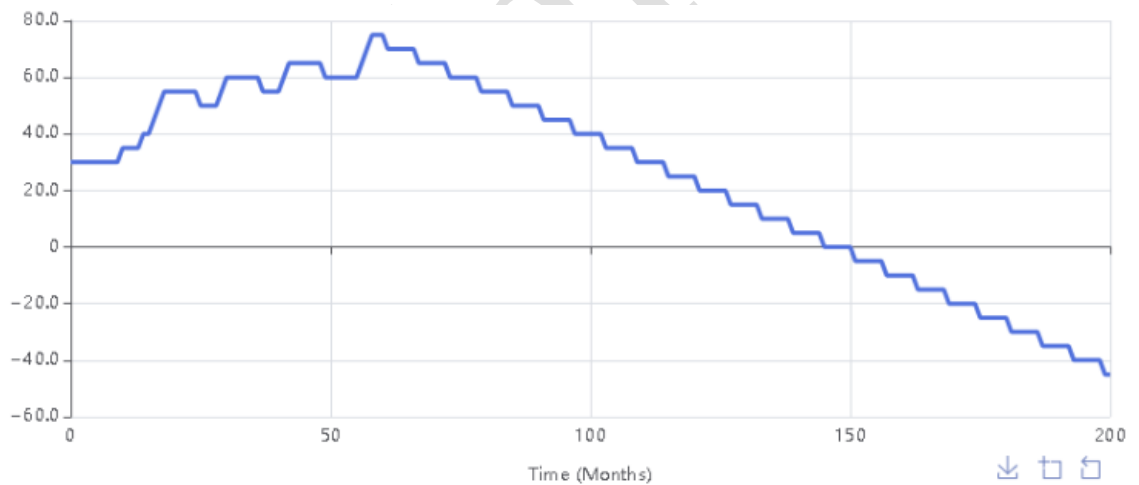
Time	Capacity	Details warehouse	Demand	Sale of details	Investments in capaci	Dividend payments
0	30	0	0	0	0	0
1	30	291.6	85	85	0	0
2	30	498.2	170	170	2,198	942
3	30	571.2	255	255	8,148	3,492
4	30	559.2	340	340	14,098	6,042
5	30	462.2	425	425	20,048	8,592
6	30	280.2	510	280.2	4,312	1,848
7	30	243	595	243	5,367.6	2,300.4
8	30	291.6	680	291.6	8,769.6	3,758.4
9	30	291.6	765	291.6	3,169.6	1,358.4
10	35	291.6	850	291.6	5,109.3	2,189.7
11	35	463.05	935	463.05	17,110.8	7,333.2
12	35	463.05	1,020	463.05	11,510.8	4,933.2
13	35	463.05	1,105	463.05	7,459.76	5,861.24
14	40	529.2	1,190	529.2	11,854.08	9,313.92
15	40	777.6	1,275	777.6	21,284.48	16,723.52

Figure 4. Dynamics of investment and dividend with erosion rate = 14%



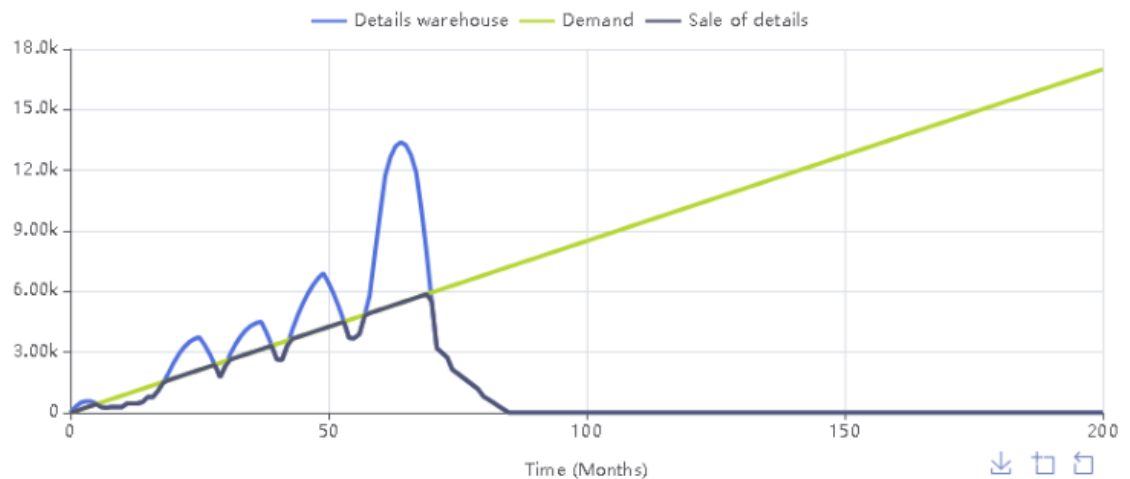
At week 61, all profit goes to dividends, but by week 85 profit becomes zero, so dividends stop.

Figure 5. Dynamics of capacity with erosion rate = 14%



Capacity grows during the first 60 weeks, and then declines every 6 months (equipment breakdown frequency). Capacity reaches zero at week 145. Although production no longer generates profit, some machines can still be sold until they break down completely.

Figure 6. Dynamics of warehouse, sales and demand with erosion rate = 14%



Demand grows steadily, but sales fluctuate. Sales are sometimes limited by demand, sometimes by supply (available stock). If the warehouse is low, the company sells whatever is available. If production cannot increase output due to insufficient capacity and workers, sales gradually drop to zero. Here, sales drop to zero at week 85, when the last worker is laid off because there is no money for investments.

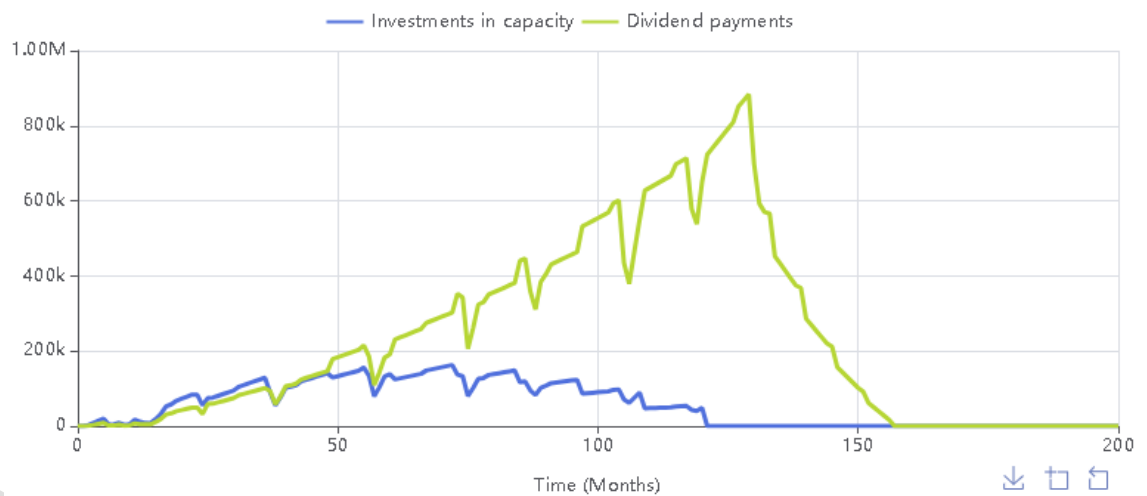
2.2. Experiment with erosion rate = 7%

The results of calculations of the model in Insight Maker with an erosion of shares equal to 7% for the first 15 steps are shown in Figure 7. Graphs of the dynamics of investments in capacity and dividend payments (Figure 8), dynamics of capacity (Figure 9), dynamics of warehouse, sales and demand (Figure 10) for the entire time are also presented (200 steps).

Figure 7. Model results with erosion rate = 7%

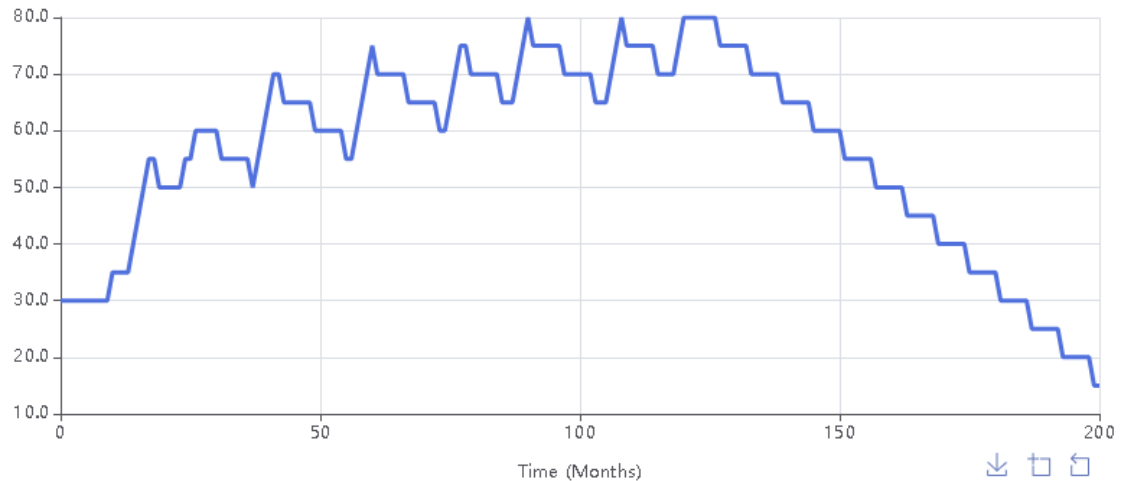
Time	Capacity	Details warehouse	Demand	Sale of details	Investments in capaci	Dividend payments
0	30	0	0	0	0	0
1	30	291.6	85	85	0	0
2	30	498.2	170	170	2,198	942
3	30	571.2	255	255	8,148	3,492
4	30	559.2	340	340	14,098	6,042
5	30	462.2	425	425	20,048	8,592
6	30	280.2	510	280.2	4,312	1,848
7	30	243	595	243	5,367.6	2,300.4
8	30	291.6	680	291.6	8,769.6	3,758.4
9	30	291.6	765	291.6	3,169.6	1,358.4
10	35	291.6	850	291.6	5,109.3	2,189.7
11	35	463.05	935	463.05	17,110.8	7,333.2
12	35	463.05	1,020	463.05	11,510.8	4,933.2
13	35	463.05	1,105	463.05	8,392.23	4,928.77
14	40	529.2	1,190	529.2	8,295.84	4,872.16
15	45	777.6	1,275	777.6	18,830.7	11,059.3

Figure 8. Dynamics of investment and dividend with erosion rate = 7%



The share of profit to investments reaches zero at week 121. Profit itself (which had been fully distributed as dividends) reaches zero at week 157.

Figure 9. Dynamics of capacity with erosion rate = 7%



Capacity grows steadily until week 126, and then declines due to the inability to purchase new equipment that would increase capacity.

Figure 10. Dynamics of warehouse, sales and demand with erosion rate = 7%



Demand grows steadily, but the company cannot increase output because capacity is falling. Sales drop to zero at week 157.

2.3. Experiment with erosion rate = 0%

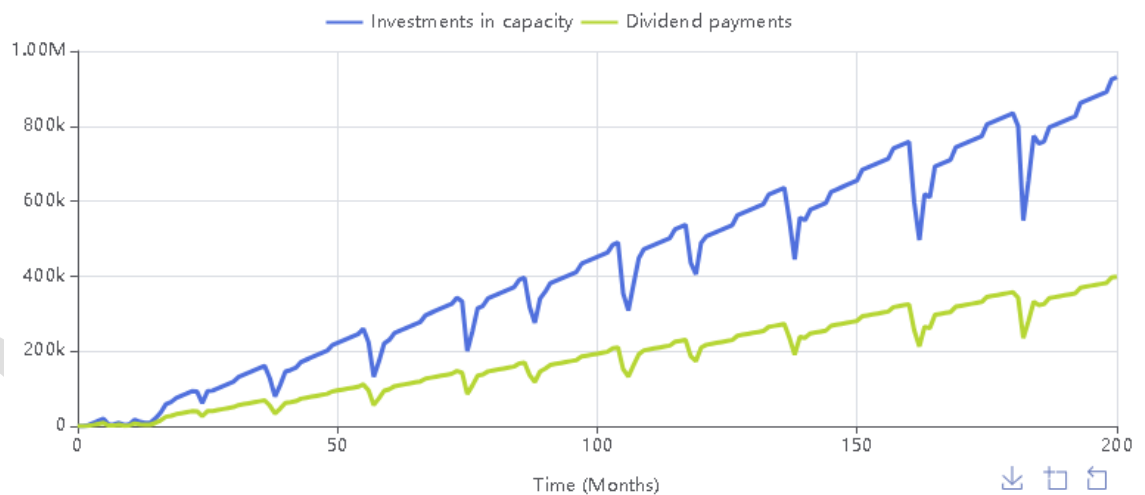
The results of calculations of the model in Insight Maker with an erosion of shares equal to 0% for the first 15 steps are shown in Figure 11. Graphs of the dynamics of investments in capacity and dividend payments (Figure 12),

dynamics of capacity (Figure 13), dynamics of warehouse, sales and demand (Figure 14) for the entire time are also presented (200 steps).

Figure 7. Model results with erosion rate = 0%

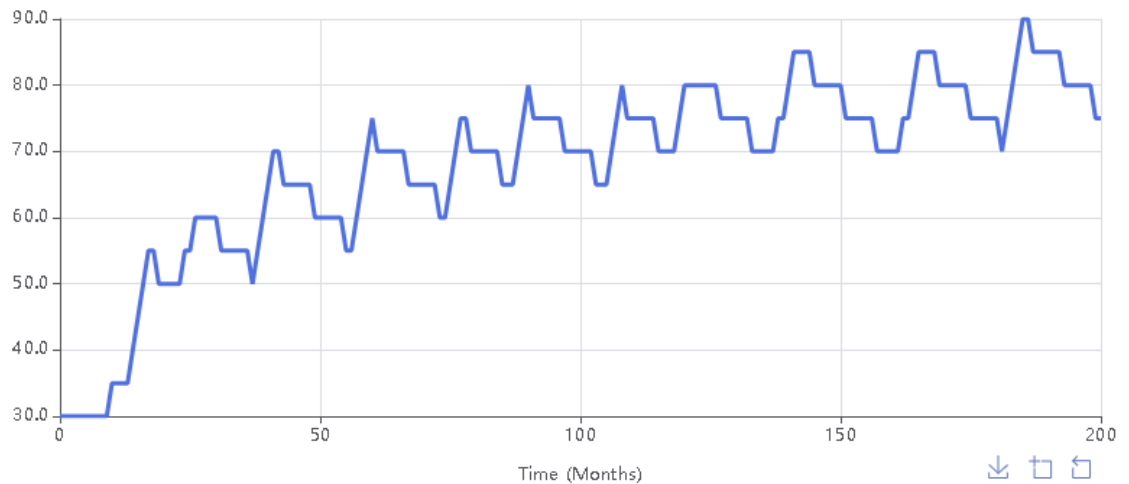
Time	Capacity	Details warehouse	Demand	Sale of details	Investments in capaci	Dividend payments
0	30	0	0	0	0	0
1	30	291.6	85	85	0	0
2	30	498.2	170	170	2,198	942
3	30	571.2	255	255	8,148	3,492
4	30	559.2	340	340	14,098	6,042
5	30	462.2	425	425	20,048	8,592
6	30	280.2	510	280.2	4,312	1,848
7	30	243	595	243	5,367.6	2,300.4
8	30	291.6	680	291.6	8,769.6	3,758.4
9	30	291.6	765	291.6	3,169.6	1,358.4
10	35	291.6	850	291.6	5,109.3	2,189.7
11	35	463.05	935	463.05	17,110.8	7,333.2
12	35	463.05	1,020	463.05	11,510.8	4,933.2
13	35	463.05	1,105	463.05	9,324.7	3,996.3
14	40	529.2	1,190	529.2	9,217.6	3,950.4
15	45	777.6	1,275	777.6	20,923	8,967

Figure 12. Dynamics of investment and dividend with erosion rate = 0%



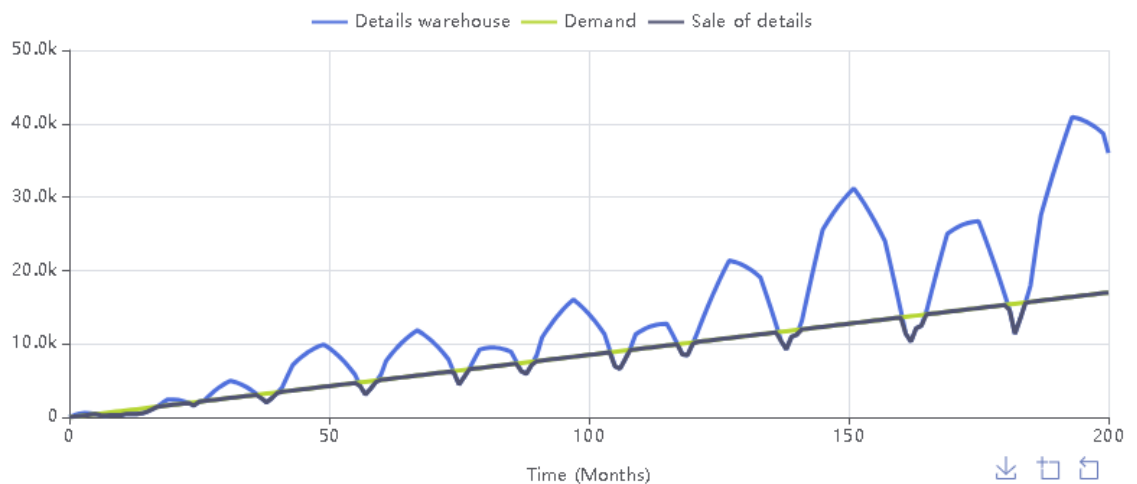
With fixed shares (70% investment, 30% dividends), both investments and dividends grow steadily.

Figure 13. Dynamics of capacity with erosion rate = 0%



Capacity also grows steadily throughout the simulation. The ups and downs correspond to equipment purchases (ups) and periods when the company produces more than demand, so broken machines are not replaced until necessary.

Figure 14. Dynamics of warehouse, sales and demand with erosion rate = 0%



In this scenario, the company can meet customer demand by producing the required number of parts. The warehouse stock also shows ups and downs linked to production capacity.

3. Results

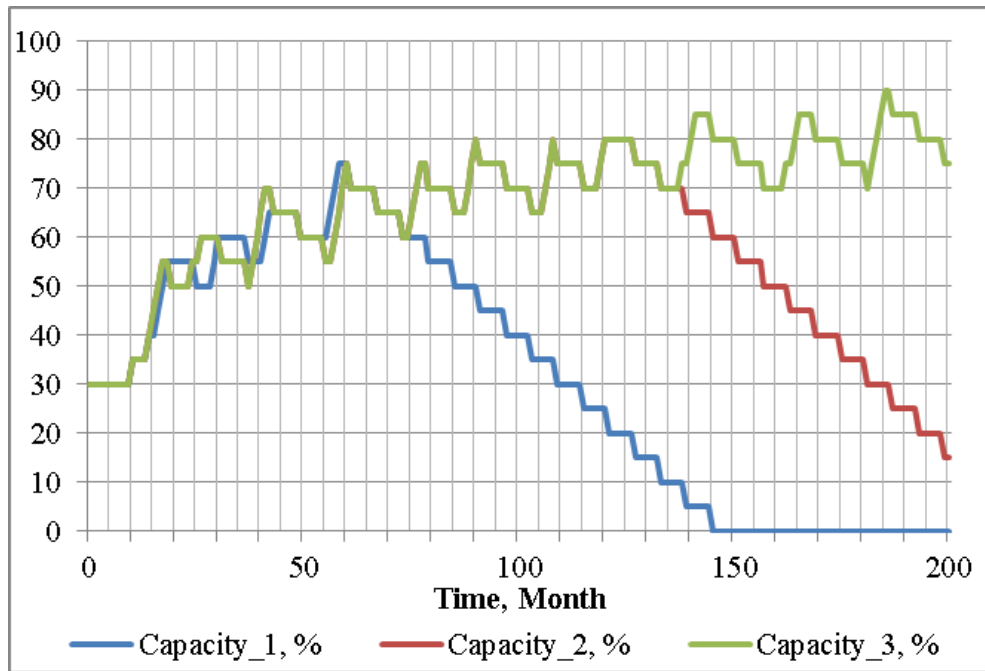
For convenience, all calculations were transferred to Excel, along with a summary table presented in Figure 15 for the first 40 steps. Below are also

comparative graphs of capacity dynamics (Figure 16), sales (Figure 17), investments to capacity (Figure 18), and dividend payments (Figure 19) for the entire time period.

Figure 15. Summary table for all three experiments

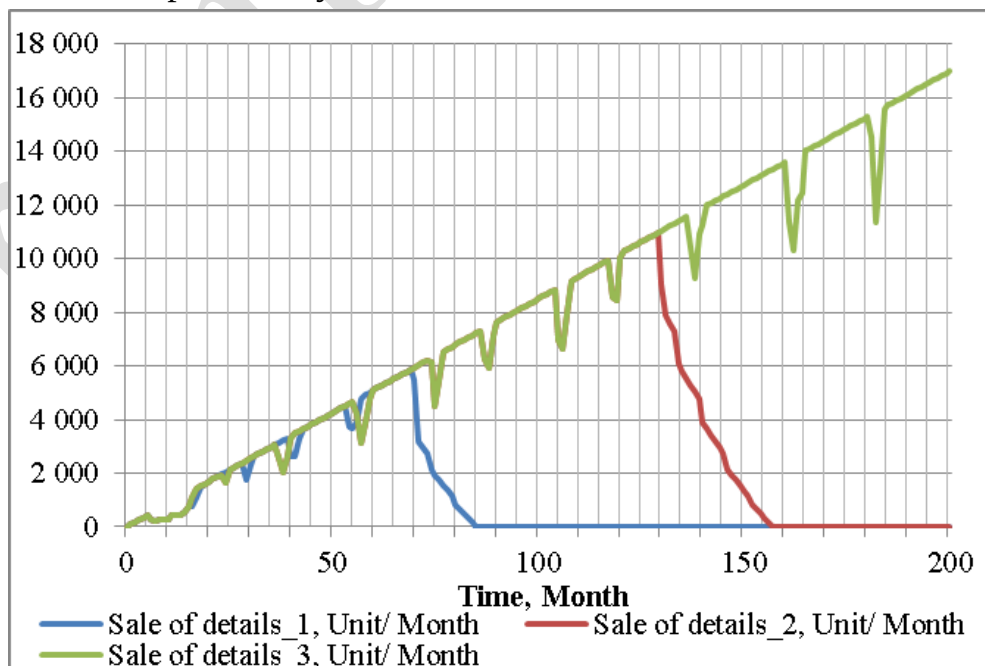
	Capacity, %			Sale of details, Unit/ Month			Investments in capacity, \$			Dividend payments, \$		
	Mod_1/	Mod_2	Mod_3	Mod_1/	Mod_2	Mod_3	Mod_1/	Mod_2	Mod_3	Mod_1/	Mod_2	Mod_3
0	30	30	30	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00
1	30	30	30	85,00	85,00	85,00	0,00	0,00	0,00	0,00	0,00	0,00
2	30	30	30	170,00	170,00	170,00	2 198,00	2 198,00	2 198,00	942,00	942,00	942,00
3	30	30	30	255,00	255,00	255,00	8 148,00	8 148,00	8 148,00	3 492,00	3 492,00	3 492,00
4	30	30	30	340,00	340,00	340,00	14 098,00	14 098,00	14 098,00	6 042,00	6 042,00	6 042,00
5	30	30	30	425,00	425,00	425,00	20 048,00	20 048,00	20 048,00	8 592,00	8 592,00	8 592,00
6	30	30	30	280,20	280,20	280,20	4 312,00	4 312,00	4 312,00	1 848,00	1 848,00	1 848,00
7	30	30	30	243,00	243,00	243,00	5 367,60	5 367,60	5 367,60	2 300,40	2 300,40	2 300,40
8	30	30	30	291,60	291,60	291,60	8 769,60	8 769,60	8 769,60	3 758,40	3 758,40	3 758,40
9	30	30	30	291,60	291,60	291,60	3 169,60	3 169,60	3 169,60	1 358,40	1 358,40	1 358,40
10	35	35	35	291,60	291,60	291,60	5 109,30	5 109,30	5 109,30	2 189,70	2 189,70	2 189,70
11	35	35	35	463,05	463,05	463,05	17 110,80	17 110,80	17 110,80	7 333,20	7 333,20	7 333,20
12	35	35	35	463,05	463,05	463,05	11 510,80	11 510,80	11 510,80	4 933,20	4 933,20	4 933,20
13	35	35	35	463,05	463,05	463,05	7 459,76	8 392,23	9 324,70	5 861,24	4 928,77	3 996,30
14	40	40	40	529,20	529,20	529,20	11 854,08	8 295,84	9 217,60	9 313,92	4 872,16	3 950,40
15	40	45	45	777,60	777,60	777,60	21 284,48	18 830,70	20 923,00	16 723,52	11 039,30	8 967,00
16	45	50	50	777,60	1 093,50	1 093,50	16 738,40	32 665,50	36 295,00	13 151,60	19 184,50	15 555,00
17	50	55	55	1 093,50	1 445,00	1 445,00	29 036,00	52 728,48	58 587,20	22 814,00	30 967,52	25 108,80
18	55	55	55	1 483,00	1 530,00	1 530,00	44 629,76	58 083,48	64 537,20	35 066,24	34 112,52	27 658,80
19	55	50	50	1 615,00	1 615,00	1 615,00	53 552,24	67 725,00	75 250,00	42 076,76	39 775,00	32 250,00
20	55	50	50	1 700,00	1 700,00	1 700,00	58 312,24	73 080,00	81 200,00	45 816,76	42 920,00	34 800,00
21	55	50	50	1 785,00	1 785,00	1 785,00	63 072,24	78 435,00	87 150,00	49 556,76	46 065,00	37 350,00
22	55	50	50	1 870,00	1 870,00	1 870,00	67 832,24	83 790,00	93 100,00	53 296,76	49 210,00	39 900,00
23	55	50	50	1 955,00	1 940,40	1 940,40	72 592,24	83 185,20	92 428,00	57 036,76	48 854,80	39 612,00
24	55	55	55	2 040,00	1 620,00	1 620,00	77 352,24	55 521,27	61 690,30	60 776,76	32 607,73	26 438,70
25	50	55	55	2 125,00	2 123,55	2 123,55	64 680,00	74 713,52	93 391,90	89 320,00	58 703,48	40 025,10
26	50	60	60	2 210,00	2 210,00	2 210,00	68 250,00	75 980,80	94 976,00	94 250,00	59 699,20	40 704,00
27	50	60	60	2 295,00	2 295,00	2 295,00	71 820,00	80 740,80	100 926,00	99 180,00	63 439,20	43 254,00
28	50	60	60	2 371,50	2 380,00	2 380,00	71 673,00	85 500,80	106 876,00	98 977,00	67 179,20	45 804,00
29	55	60	60	1 755,00	2 465,00	2 465,00	40 556,04	90 260,80	112 826,00	56 005,96	70 919,20	48 354,00
30	60	60	60	2 286,90	2 550,00	2 550,00	56 855,40	95 020,80	118 776,00	78 514,60	74 659,20	50 904,00
31	60	55	55	2 635,00	2 635,00	2 635,00	72 446,64	104 997,20	131 246,50	100 045,36	82 497,80	56 248,50
32	60	55	55	2 720,00	2 720,00	2 720,00	76 016,64	109 757,20	137 196,50	104 975,36	86 237,80	58 798,50
33	60	55	55	2 805,00	2 805,00	2 805,00	79 586,64	114 517,20	143 146,50	109 905,36	89 977,80	61 348,50
34	60	55	55	2 890,00	2 890,00	2 890,00	83 156,64	119 277,20	149 096,50	114 835,36	93 717,80	63 898,50
35	60	55	55	2 975,00	2 975,00	2 975,00	86 726,64	124 037,20	155 046,50	119 765,36	97 457,80	66 448,50
36	60	55	55	3 060,00	3 060,00	3 060,00	90 296,64	128 797,20	160 996,50	124 695,36	101 197,80	68 998,50
37	55	50	50	3 145,00	2 583,40	2 583,40	65 359,84	89 591,60	127 988,00	168 068,16	93 248,40	54 852,00
38	55	55	55	3 230,00	2 025,00	2 025,00	67 739,84	55 579,72	79 399,60	174 188,16	57 848,28	34 028,40
39	55	60	60	3 315,00	2 613,60	2 613,60	70 119,84	76 765,36	109 664,80	180 308,16	79 898,64	46 999,20

Figure 16. Comparative dynamics of capacity



The only positive scenario for the company is Experiment 3 (zero erosion). Experiments 1 and 2 inevitably drive the company's main asset — production capacity — to zero. It is only a matter of time: even 1% erosion would lead to collapse.

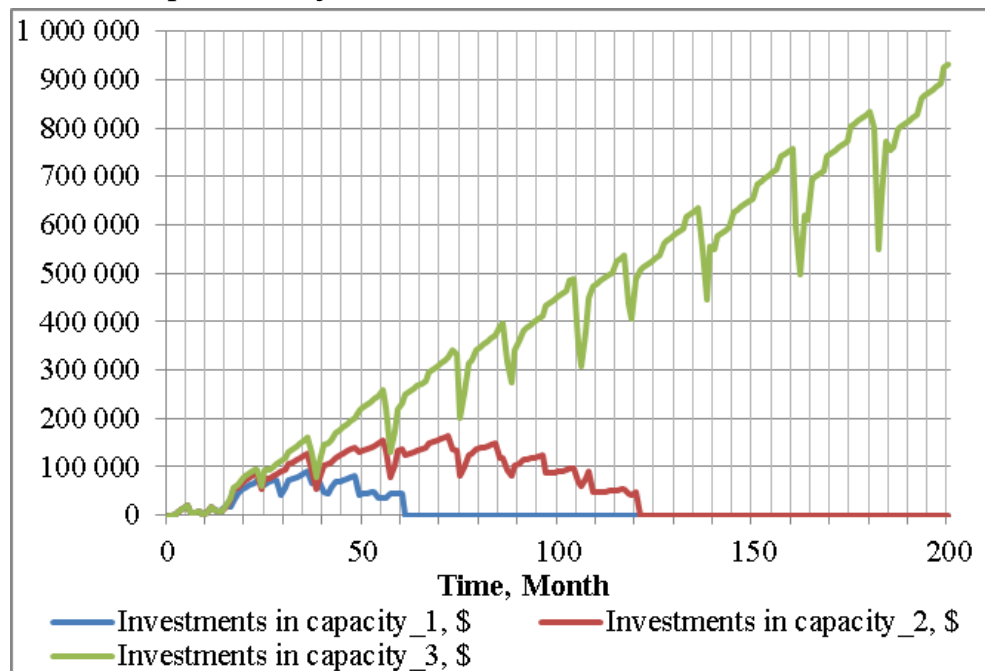
Figure 17. Comparative dynamics of sales



Scenario 3 favorably affects investment: the more the company produces, the more it sells, the more profit remains for investment. Experiments 1 and 2 lead

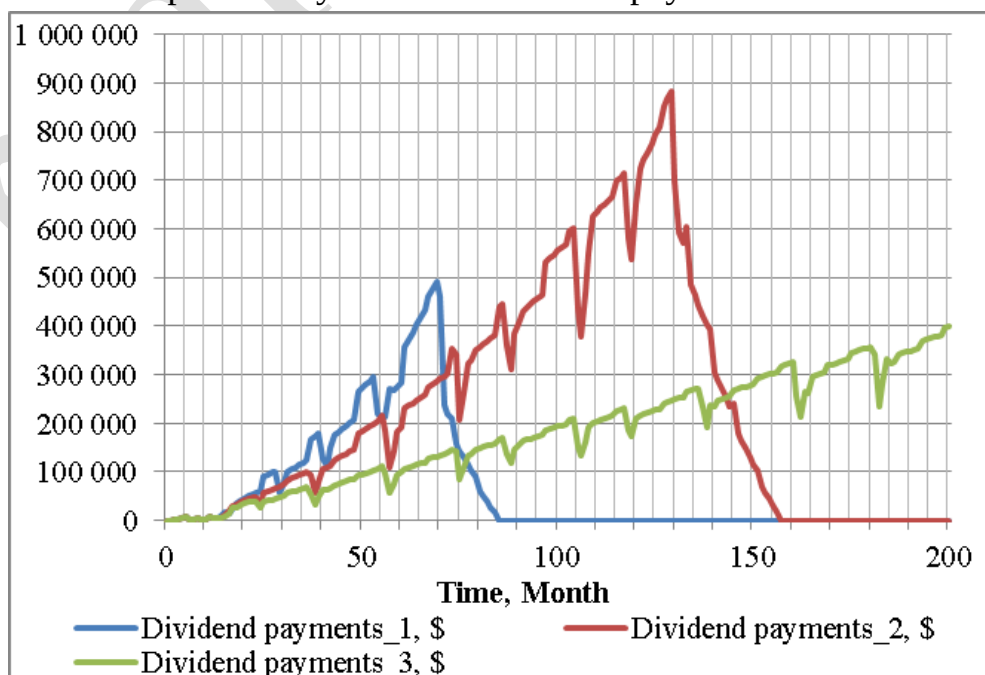
to zero sales — the company can neither pay its workers nor invest in new equipment.

Figure 18. Comparative dynamics of investment



Experiments 1 and 2 show that any shift of profit shares toward dividends leads to collapse. Experiment 3 (zero erosion) allows the company to grow its profit, and with it, both investment and dividend payments.

Figure 19. Comparative dynamics of dividend payments



Experiment 2 looks attractive in terms of total dividend payouts, but their fall to zero bodes nothing good. In Experiment 3, dividends are lower, but they are regular and grow over time. If the company continues to invest, dividend payouts could reach the levels of Experiment 2 while maintaining the regularity of Experiment 3.

4. Conclusion

This work examined the "Growth and Underinvestment" archetype under three scenarios: rapid erosion (14%), moderate erosion (7%), and zero erosion (0%). The main findings are:

1. Any positive erosion rate leads to collapse. At 14%, the company reaches zero in 85 weeks; at 7%, in 157 weeks. The difference is only in time.

2. The only sustainable scenario is zero erosion. With fixed shares (70% investment, 30% dividends), the company grows steadily: capacity increases, sales rise, dividends are regular and increasing.

3. Dividend size is deceptive. The 7% erosion scenario looks attractive in terms of total payouts, but their fall to zero makes that growth illusory. In the zero-erosion scenario, dividends are lower, but they are sustainable and grow over time.

Thus, the "Growth and Underinvestment" archetype demonstrates that even a small but systematic shift of profit from investment to current consumption (dividends) inevitably leads to degradation and collapse. Sustainable growth is possible only when a long-term balance is maintained.

Acknowledgments

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